Highlights

When Do Dendritic Artificial Neural Networks Help? A Controlled Benchmark Study with Parameter, Branching, and Timing Controls

Özkan Canay

- The study benchmarks dendritic artificial neural networks under parameter matching, naive branching control, reduced-data diagnostics, and CPU timing analysis.
- Manuscript tables use test accuracy at the best validation epoch, reconstructed from per-epoch histories rather than historical best-test exports.
- DANN-LRF is strongest among low-parameter models on Fashion-MNIST and CIFAR-10 full, while DANN-RANDOM is numerically preferred on KMNIST.
- Reduced-dataset benefits are mixed, and the extra CPU cost of dendritic computation beyond naive branching is small.

When Do Dendritic Artificial Neural Networks Help? A Controlled Benchmark Study with Parameter, Branching, and Timing Controls

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Abstract

Dendritic artificial neural networks (DANNs) introduce branch-level local computation before somatic integration as a biologically inspired alternative to flat artificial neurons. What remains unclear is whether their empirical gains come from dendritic nonlinear integration itself, from branching alone, or from unconstrained capacity differences. This study presents a controlled benchmark centered on that question. We evaluate three DANN sampling strategies, namely local receptive fields (DANN-LRF), random sampling (DANN-RANDOM), and grouped receptive fields (DANN-GRF), against a parameter-matched multilayer perceptron (MLP-Param), a naive branching control with identical routing but no dendrite-level nonlinearity (Naive-Branch), and a higher-capacity dense reference (VANN-Same). The benchmark covers FashionMNIST, KMNIST, and CIFAR-10, together with historical reduced-dataset diagnostic runs and separate CPU-only timing runs. To avoid test-set selection bias, the manuscript reconstructs the primary accuracy tables from per-epoch histories and reports test accuracy at the best validation epoch. Across the three full-data conditions, DANN-LRF consistently outperforms MLP-Param, with mean gains of 1.67 percentage points on FashionMNIST, 4.06 points on KMNIST, and 17.04 points on CIFAR-10 under paired Wilcoxon tests with Holm correction. DANN-LRF also improves over Naive-Branch on FashionMNIST full and CIFAR-10 full,

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whereas the gap disappears on KMNIST and under the most reduced Fashion-MNIST setting. The best dendritic sampling strategy is dataset-dependent: DANN-LRF is favored on FashionMNIST, while DANN-RANDOM is numerically preferred on KMNIST. The historical reduced-dataset results are mixed and should be interpreted as diagnostics rather than strict fixed-test low-training-data evidence. CPU timing shows that DANN-LRF is slower than MLP-Param but adds only about 1% to 4% runtime beyond Naive-Branch. These findings position DANNs as parameter-efficient classifiers whose benefits depend on dataset structure and baseline discipline rather than as universal accuracy maximizers.

Keywords: dendritic artificial neural networks, parameter-efficient classification, controlled benchmark, local receptive fields, biologically inspired computation, image classification

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1. Introduction

Standard artificial neurons compress all incoming features into a single weighted sum followed by a pointwise nonlinearity. Biological dendrites, by contrast, are widely modeled as compartmental, locally nonlinear, and context-dependent computational subunits whose outputs are integrated at the soma [1–7]. This view has shifted dendrites from a purely anatomical detail to a computational design principle. Recent bridge papers in computational neuroscience and machine learning argue that dendritic mechanisms are relevant not only for biological realism, but also for robust, efficient, and structured information processing in artificial systems [8–13].

Interest in dendritic inductive bias has increased for at least three reasons. First, single cortical neurons can exhibit function classes rich enough to be approximated by deep artificial networks, suggesting that branch-level structure matters computationally rather than only biologically [14]. Second, dendrite-inspired normalization and sparse connectivity studies indicate that local structure can improve learning efficiency under constrained connectivity [15–17]. Third, the broader machine learning literature has become increasingly attentive to parameter efficiency, sparse routing, and conditional computation, even when the motivation is not explicitly dendritic [18–21].

The artificial dendrite literature itself is now broad but heterogeneous. Early work emphasized nonlinear interactions within single dendritic neurons [22–27], while later studies extended the idea toward classification, forecasting,

recurrent modeling, and deeper architectures [28–41]. A parallel line has incorporated dendrites into spiking and neuromorphic systems for event-based processing, multicompartment memory, and temporally structured computation [42–50]. Dendritic ideas also intersect with biologically plausible credit assignment, predictive coding, and local learning rules [51–61].

Despite this growth, the empirical question most relevant to practical shallow classification remains under-addressed. Many papers introduce a new dendritic mechanism, evaluate on a specific application domain, or compare against conventional baselines without simultaneously controlling for parameter count, branch structure, and runtime. As a result, it is often unclear whether the observed gains come from dendritic nonlinear integration itself, from structured branching without true dendritic computation, or simply from a larger or better tuned model.

This study addresses that gap with a deliberately controlled benchmark. The main candidate method is DANN-LRF, a dendritic artificial neural network with fixed local receptive-field routing. The broader DANN family also includes DANN-RANDOM and DANN-GRF. These models are compared against three baselines chosen to answer different fairness questions: a parameter-matched multilayer perceptron (MLP-Param), a naive branching control with identical routing but without dendrite-level nonlinearity (Naive-Branch), and a much larger dense reference (VANN-Same). The benchmark covers FashionMNIST [62], KMNIST [63], and CIFAR-10 [64], together with reduced-data diagnostics and CPU timing runs.

The manuscript is intentionally not framed as a state-of-the-art image-classification paper. The goal is narrower and, for this architecture family, more informative: to determine whether dendritic computation improves parameter-efficient classification under matched baselines, to identify when local receptive-field routing helps, and to quantify the runtime cost of those gains.

The paper makes four contributions.

1. It provides a controlled empirical benchmark of the DANN family under parameter matching, branching controls, reduced-data diagnostics, and timing analysis.
2. It reconstructs the primary accuracy evidence from per-epoch histories and reports test accuracy at the best validation epoch, rather than relying on historical best-test exports.
3. It shows that DANN-LRF consistently outperforms the parameter-

matched MLP on the full-data benchmarks and on CIFAR-10 at 20% archived subset size, while the advantage over naive branching is smaller and condition-dependent.

4. It shows that the preferred dendritic sampling strategy is dataset-dependent, with local receptive fields favored on FashionMNIST and random sampling numerically favored on KMNIST.

2. Related Work and Study Positioning

The recent literature can be organized into four partially overlapping groups. The first group provides biological and conceptual motivation. Reviews and perspectives emphasize dendrites as multistage nonlinear processors, link dendritic properties to efficiency and memory formation, and argue that the neuroscience literature contains actionable architectural ideas for machine learning [1–13]. These papers justify the search for dendrite-inspired networks, but they do not establish the benchmark conditions under which such networks help.

The second group develops dendrite-inspired artificial neural architectures directly. In this line, the field has moved from single-neuron nonlinear interaction models [22, 23, 25–27] to broader network variants for classification, prediction, and forecasting [28–41]. This literature clearly demonstrates sustained interest in dendritic models, but much of it emphasizes task-wise improvement over selected baselines rather than benchmark discipline across parameter budgets, branching controls, and timing.

The third group extends dendritic ideas into spiking, neuromorphic, and temporal processing settings. Representative examples include dendritic event-based learning, self-adaptive multicompartment spiking neurons, multicompartment neuromorphic learning systems, Dendrify, temporal dendritic heterogeneity for multi-timescale dynamics, and dendritic hardware-oriented architectures [42–50, 65]. These studies are important because they show that dendritic computation is not a niche ANN idea. However, their objectives are usually different from ours, focusing on neuromorphic learning, temporal memory, or hardware realization rather than shallow parameter-aware image classification.

The fourth group concerns biologically plausible learning and structured efficiency more broadly. Segregated dendrites, predictive coding, feedback control, target propagation, localized learning, and continual-learning mechanisms have all been studied as alternatives or complements to standard

backpropagation [51–61, 66]. In parallel, sparse and structured connectivity work has highlighted the importance of routing and normalization in efficient learning [15–21]. This adjacent literature reinforces our framing: the practical question is not whether dendritic inspiration is interesting, but whether it produces measurable value once fairness constraints are imposed.

Table 1 summarizes representative recent studies that are close to the present manuscript in spirit. The table also makes the remaining niche of the current work explicit: a controlled, shallow, parameter-aware benchmark that separates dendritic nonlinearity from branching structure.

Table 1: Representative recent studies on dendritic or dendrite-inspired machine learning models.

Study	**Model family**	**Main setting**	**Gap vs. present study**
Bird et al. [15]	Sparse ANN normalization	Dendritic normalization for sparsely connected ANNs	Normalization focus rather than branch-versus-nonlinearity controls or multi-dataset shallow benchmarking.
Yang et al. [42]	Spiking / event-based	Dendritic event-based learning in spiking networks	Neuromorphic setting rather than a parameter-matched ANN classifier benchmark.
Acharya et al. [10]	Review / perspective	Broad survey of dendritic computing for machine learning	Motivates the area but does not provide controlled comparative evidence.
Egrioglu et al. [32]	Recurrent DNM	Time-series forecasting with recurrent dendritic neurons	Different task family and no naive branching control.
Yang et al. [43]	Multicompartment SNN	Working-memory-oriented spiking neuron model	Temporal spiking focus rather than shallow image classification.
Yilmaz and Yolcu [33]	DNM network	Robust training for time-series prediction	Strong forecasting focus without matched dense and branched controls.
Pagkalos et al. [45]	Spiking framework	Software framework for adding dendrites to SNNs	Framework contribution, not a controlled ANN benchmark.
Tang et al. [37]	Dendritic neural network	Extension of DNM toward higher-dimensional classification tasks	Does not isolate branching alone or report matched CPU timing.
Egrioglu and Bas [38]	Deep DANN	Deep dendritic ANN for forecasting	Deep forecasting architecture, not a shallow image benchmark with control triads.
Li et al. [39]	Dendritic classifier	Alternating excitation-inhibition dendritic computation for classification	Classification-oriented, but not built around parameter- and branch-matched fairness constraints.
Liu et al. [40]	Dendrite-inspired ANN	Correlation-sensitive artificial network design	Architecturally close to modern ANN framing, but not centered on the baseline-isolation question studied here.
Du et al. [67]	Lightweight dendritic CNN	Dendritic ShuffleNet for medical image classification	Application study rather than a shallow benchmark analysis.
Chavlis and Poirazi [13]	Bio-inspired ANN	Parameter-efficient and robust learning with dendritic mechanisms	Strong evidence for dendritic value, but with a different architecture and without the present branching-control design.

This positioning leads to a deliberately modest novelty claim. The present work does not claim biological proof, state-of-the-art vision performance, or universal superiority of a single DANN variant. Instead, it offers a controlled answer to a narrower question: when fixed dendritic routing and local nonlinear integration help under matched shallow baselines, when routing strategy matters, and how much extra CPU cost the nonlinear dendritic stage adds beyond branching alone.

3. Materials and Methods

3.1. Study design

Figure 1 summarizes the study logic. The scientific question is whether dendritic architectures provide measurable value under constrained parameter budgets. The benchmark therefore combines fairness controls (parameter matching and naive branching), three full-data datasets, archived reduced-data diagnostics, and separate CPU timing runs. The two primary contrasts are DANN-LRF versus MLP-Param and DANN-LRF versus Naive-Branch.

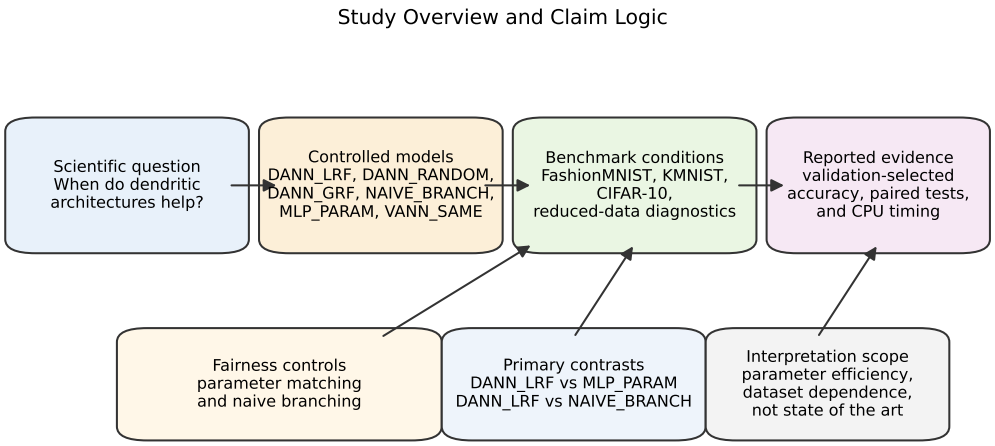

Figure 1: Study overview. The benchmark is designed around controlled contrasts rather than around absolute leaderboard performance. The main evidence combines validation-selected accuracy, seed-paired statistics, and CPU timing.

3.2. DANN layer, routing strategies, and controls

Figure 2 illustrates the layer logic shared by the DANN family. The formulation follows the general dendritic neuron modeling idea of local branch processing followed by soma-level aggregation [22, 23, 37, 38], but is intentionally simplified to support controlled benchmarking. Let $x \in \mathbb{R}^D$ denote a flattened input, S the number of soma units, B the number of branches per soma, $N = SB$ the number of dendrites, and K the number of sampled features per dendrite. The model keeps a fixed routing tensor $I \in \mathbb{N}^{N \times K}$, sampled once at initialization and held fixed during training. Dendrite n receives the subvector $x[I_n]$ and applies a local linear map followed by LeakyReLU:

$$z_n = \sum_{k=1}^{K} W_{n,k} x[I_{n,k}] + b_n, \tag{1}$$

$$a_n = \mathrm{LeakyReLU}_\alpha(z_n). \tag{2}$$

The B dendrites that belong to soma s are then combined through cable weights:

$$u_s = \sum_{b=1}^{B} C^c_{s,b}\, a_{sB+b} + b^s_s, \tag{3}$$

$$h_s = \mathrm{LeakyReLU}_\alpha(u_s), \tag{4}$$

$$y = W^o h + b^o. \tag{5}$$

Three routing strategies are evaluated.

- DANN-LRF: each dendrite samples $K = 16$ features from a randomly placed 4×4 local patch. In CIFAR-10, the patch spans all three channels.
- DANN-RANDOM: each dendrite samples $K = 16$ features uniformly from the full flattened input.
- DANN-GRF: all branches that feed the same soma share the same local patch, but each branch samples its own subset within that patch.

The baselines are chosen to isolate different explanations.

- Naive-Branch uses the same routed branch structure as DANN-LRF but removes dendrite-level nonlinearity before the soma stage.

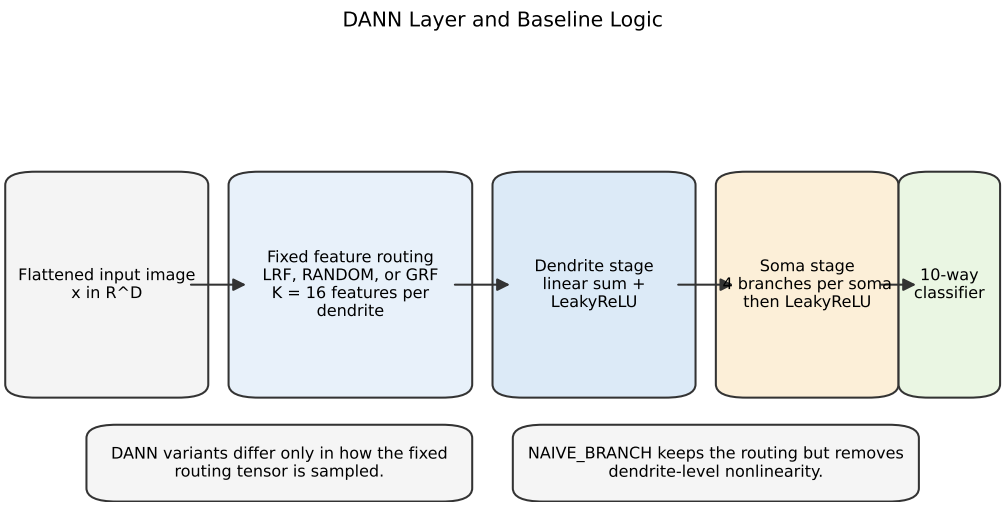

Figure 2: DANN layer and baseline logic. DANN variants differ only in how the fixed routing tensor is sampled. Naive-Branch keeps the same routed branch structure but removes the dendrite-level nonlinearity.

- MLP-Param is a two-hidden-layer multilayer perceptron whose width is selected to minimize parameter-count mismatch relative to the DANN model.
- VANN-Same is a wider two-hidden-layer dense network that serves as a high-capacity reference rather than as a parameter-matched baseline.

3.3. Datasets, conditions, and archived scope

The study uses FashionMNIST, KMNIST, and CIFAR-10 as flattened-input ten-class image-classification tasks. Six archived accuracy conditions are available in the historical result package:

- FashionMNIST full
- KMNIST full
- CIFAR-10 full
- FashionMNIST 0.2
- FashionMNIST 0.1

- CIFAR-10 0.2

The full-data FashionMNIST and KMNIST runs include all six low-parameter models plus VANN-Same. The archived CIFAR-10 folders include DANN-LRF, Naive-Branch, MLP-Param, and VANN-Same. The reduced-data folders focus on the main candidate method and its main controls; the archived CIFAR-10 0.2 folder also includes VANN-Same.

An important historical detail is that the archived reduced-data runs were produced before the repository cleanup that separated training and test subsetting. In those historical folders, the same subset-fraction logic also affected the test loader. For that reason, the reduced-data results are treated in this manuscript as reduced-dataset diagnostics rather than as strict fixed-test low-training-data robustness experiments.

3.4. Training protocol and parameter budgets

Table 2 summarizes the shared hyperparameters and parameter counts. Unless otherwise noted, accuracy runs used 30 epochs, Adam [68] with learning rate 10^{-3}, batch size 256, and seeds $\{0, \ldots, 9\}$. Timing runs used 10 epochs, seeds $\{0, 1, 2\}$, and CPU-only execution. The archived accuracy runs were completed across mixed hardware contexts because they were collected in separate stages, but timing is reported only from the dedicated CPU protocol.

The FashionMNIST and KMNIST parameter mismatch between DANN and MLP-Param is about 1%. On CIFAR-10 the mismatch is larger, about 14.7% relative to MLP-Param, and should be kept in mind when interpreting parameter-efficiency claims on that dataset.

3.5. Primary metric, paired statistics, and sensitivity audit

The historical summary CSV files export `best_test_acc`, which is the maximum test accuracy observed across epochs. Because that metric is test-selected, it is not used as the primary manuscript evidence. Instead, the manuscript reconstructs a validation-selected metric directly from the per-epoch history files: for each seed and model, we identify the epoch with the best validation loss and then report the test accuracy at that epoch. All tables and paired tests in the manuscript use this reconstructed metric.

For each condition, we computed paired Wilcoxon signed-rank tests [69] on seed-level validation-selected accuracies for the contrasts DANN-LRF versus MLP-Param, DANN-LRF versus Naive-Branch, and DANN-LRF versus DANN-RANDOM where available. Holm correction [70] was

Table 2: Shared hyperparameters and parameter budgets.

Setting	**Value**
Soma units S	128
Branches per soma B	4
Dendrites $N = SB$	512
Sampled features per dendrite K	16
LRF/GRF patch size	4×4
Activation	LeakyReLU, slope 0.1
Optimizer	Adam
Learning rate	10^{-3}
Loss	Cross-entropy
Batch size	256
Epochs (accuracy runs)	30
Epochs (timing runs)	10
Accuracy seeds	$\{0, \ldots, 9\}$
Timing seeds	$\{0, 1, 2\}$
DANN/NAIVE params (FashionMNIST, KMNIST)	10,634
MLP-PARAM params (FashionMNIST, KMNIST)	10,527
VANN-SAME params (FashionMNIST, KMNIST)	468,874
DANN/NAIVE params (CIFAR-10)	10,634
MLP-PARAM params (CIFAR-10)	9,271
VANN-SAME params (CIFAR-10)	1,640,330

applied within each condition. We also report bootstrap 95% confidence intervals on the mean paired difference using 10,000 resamples [71]. This choice is consistent with standard recommendations for repeated classifier comparisons that avoid over-interpreting point estimates from a single split [72]. Finally, we performed a sensitivity audit comparing condition leaders under the historical best-test export and the reconstructed validation-selected metric.

4. Results

All accuracy tables in this section report test accuracy at the best validation epoch, averaged across seeds. Historical best-test summaries are used only for the sensitivity audit discussed later. This distinction matters because it removes the main test-selection concern while preserving the available

experimental evidence.

4.1. Full-data benchmarks

Table 3 and Figure 3 summarize the three full-data conditions. The higher-capacity dense baseline, VANN-Same, achieves the highest absolute accuracy in all three tasks, which is expected given its much larger parameter budget. The more informative question for this study is how the low-parameter models compare under matched or nearly matched capacity.

Table 3: Validation-selected accuracy on the three full-data benchmarks. Accuracy is reported as mean ± standard deviation across 10 seeds.

Dataset	**Model**	**Params**	**Accuracy (%)**
FashionMNIST full	VANN-Same	468,874	88.72 ± 0.38
	DANN-LRF	10,634	87.03 ± 0.17
	DANN-GRF	10,634	86.86 ± 0.31
	Naive-Branch	10,634	86.67 ± 0.19
	DANN-RANDOM	10,634	86.48 ± 0.21
	MLP-Param	10,527	85.36 ± 0.27
KMNIST full	VANN-Same	468,874	90.01 ± 0.36
	DANN-RANDOM	10,634	81.43 ± 0.34
	Naive-Branch	10,634	80.94 ± 0.45
	DANN-LRF	10,634	80.91 ± 0.62
	DANN-GRF	10,634	78.84 ± 0.37
	MLP-Param	10,527	76.85 ± 0.25
CIFAR-10 full	VANN-Same	1,640,330	52.86 ± 0.69
	DANN-LRF	10,634	47.69 ± 0.55
	Naive-Branch	10,634	46.69 ± 0.31
	MLP-Param	9,271	30.65 ± 1.47

The strongest full-data evidence is the consistent DANN-LRF advantage over MLP-Param. On FashionMNIST full, DANN-LRF exceeds MLP-Param by 1.67 percentage points, with a bootstrap 95% confidence interval of [1.54, 1.80] and Holm-adjusted $p = 0.0059$. On CIFAR-10 full, the gap grows to 17.04 points with Holm-adjusted $p = 0.0039$. Even on KMNIST, where DANN-LRF is not the strongest DANN variant, it still exceeds MLP-Param by 4.06 points with Holm-adjusted $p = 0.0059$.

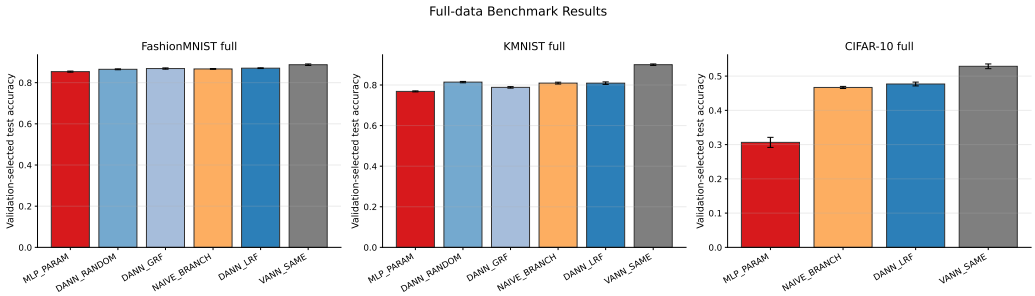

Figure 3: Validation-selected accuracy on the three full-data benchmarks. Error bars show seed-level standard deviation across 10 seeds.

The more demanding control is Naive-Branch. Here the effect is smaller and condition-dependent. DANN-LRF improves over Naive-Branch on FashionMNIST full by 0.37 points and on CIFAR-10 full by 1.01 points, both significant after Holm correction. On KMNIST, however, DANN-LRF and Naive-Branch are effectively tied, and the best DANN variant is DANN-RANDOM rather than DANN-LRF. This is a central result of the paper: dendritic computation can help, but the preferred routing strategy is not universal.

4.2. Reduced-dataset diagnostics

The historical reduced-data folders are informative, but they must be interpreted narrowly because the archival subset logic also reduced the test loader. Table 4 and Figure 4 therefore present these runs as reduced-dataset diagnostics rather than as definitive low-training-data robustness evidence.

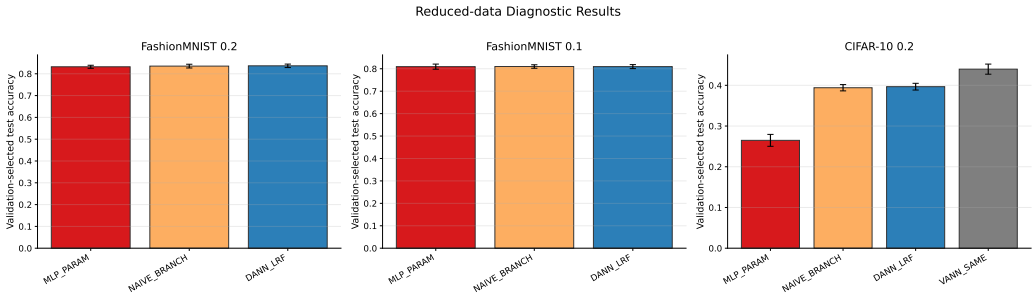

Figure 4: Validation-selected accuracy on the archived reduced-dataset diagnostic runs. Error bars show seed-level standard deviation across 10 seeds.

Table 4: Validation-selected accuracy on the archived reduced-dataset diagnostic runs. These folders were generated under historical subset logic that also affected the test loader.

Condition	**Model**	**Params**	**Accuracy (%)**
FashionMNIST 0.2	DANN-LRF	10,634	83.65 ± 0.79
	Naive-Branch	10,634	83.52 ± 0.86
	MLP-Param	10,527	83.21 ± 0.70
FashionMNIST 0.1	Naive-Branch	10,634	81.00 ± 0.78
	DANN-LRF	10,634	80.95 ± 0.89
	MLP-Param	10,527	80.94 ± 1.14
CIFAR-10 0.2	VANN-Same	1,640,330	43.97 ± 1.24
	DANN-LRF	10,634	39.66 ± 0.83
	Naive-Branch	10,634	39.41 ± 0.77
	MLP-Param	9,271	26.48 ± 1.46

The reduced-dataset diagnostics are mixed. At the 20% archived subset level, DANN-LRF remains numerically ahead of MLP-Param on both FashionMNIST and CIFAR-10, but only the CIFAR-10 gap is clearly large. FashionMNIST 0.2 shows a 0.44-point DANN-LRF advantage over MLP-Param and a 0.13-point advantage over Naive-Branch, neither statistically significant after Holm correction. FashionMNIST 0.1 is even less favorable to DANN-LRF: Naive-Branch ranks first, DANN-LRF is second, and neither contrast involving DANN-LRF is significant. In contrast, CIFAR-10 0.2 preserves a large DANN-LRF advantage over MLP-Param of 13.18 points, while the gap to Naive-Branch remains small and nonsignificant.

The evidence therefore supports a narrow reading: under the current archived reduced-dataset protocol, DANN-LRF retains some of its parameter-efficiency advantage in the moderate 20% setting, especially on CIFAR-10, but the low-data story is not strong enough to support a general robustness claim.

4.3. Paired contrasts and metric sensitivity

Table 5 collects the primary paired contrasts, and Figure 5 visualizes the same effects with bootstrap confidence intervals. The pattern is consistent with the descriptive tables: large and reliable gains against MLP-Param on full-data conditions, smaller and condition-sensitive gaps against Naive-Branch, and a dataset-dependent preference within the DANN family.

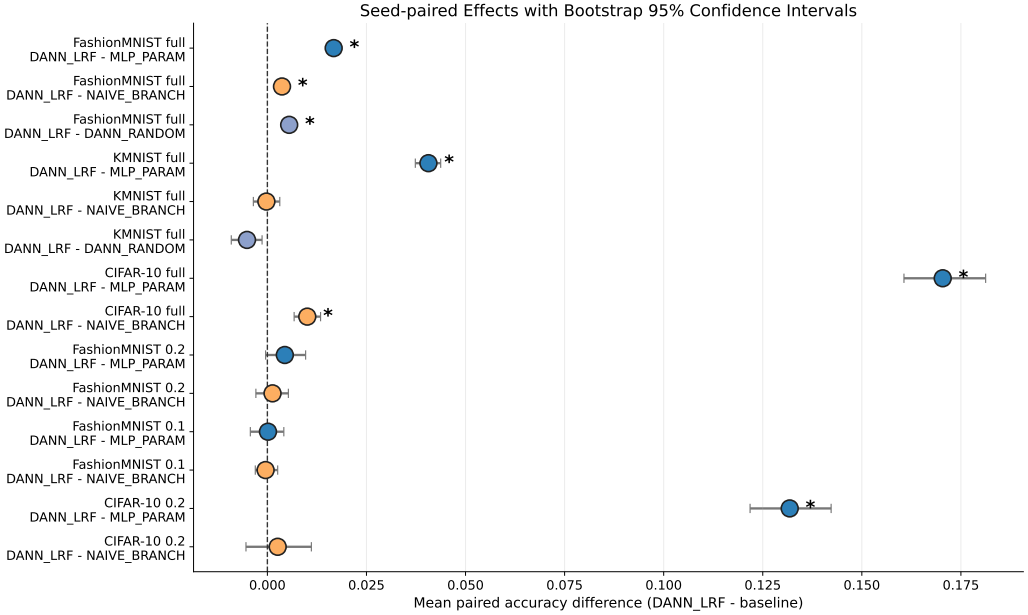

Figure 5: Seed-paired effects for DANN-LRF relative to its main baselines. Error bars denote bootstrap 95% confidence intervals on the mean paired difference. Asterisks mark contrasts with Holm-adjusted $p < 0.05$ within condition.

The sensitivity audit is reassuring. When the historical best-test exports are compared with the reconstructed validation-selected metric, the leader of each condition is preserved. Lower-order rankings do change in a few places, especially between DANN-LRF and Naive-Branch on KMNIST and between DANN-LRF and MLP-Param on FashionMNIST 0.1, but the overall full-data narrative remains unchanged. This matters because it shows that the main claims are not artifacts of using a more conservative primary metric.

4.4. CPU timing trade-offs

The timing runs were carried out separately under a CPU-only protocol. Table 6 and Figure 6 show that DANN-LRF is consistently slower than MLP-Param, but only slightly slower than Naive-Branch.

Relative to Naive-Branch, DANN-LRF adds little cost: on FashionMNIST timing the overhead is 2.95% in training and 1.02% in inference, while on CIFAR-10 timing it is 3.63% in training and 3.32% in inference. Relative to MLP-Param, however, DANN-LRF is 46.59% slower in training and

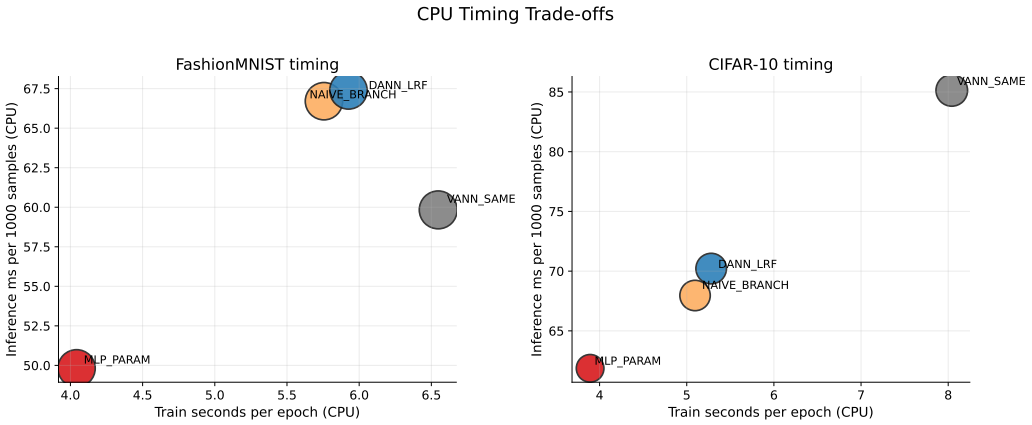

Figure 6: CPU timing trade-offs. Bubble area increases with timing-run accuracy, while the axes report training time per epoch and inference time per 1,000 samples.

35.28% slower in inference on FashionMNIST, and 35.70% slower in training and 13.50% slower in inference on CIFAR-10. The most defensible conclusion is therefore not that DANNs are cheap in absolute terms, but that the extra cost of dendritic nonlinearity beyond routed branching is modest.

5. Discussion

The central finding is that dendritic structure improves parameter-efficient classification under several, but not all, benchmark conditions. Evidence for this claim is strongest in the full-data comparisons against MLP-Param. Across FashionMNIST, KMNIST, and CIFAR-10 full, DANN-LRF consistently exceeds the parameter-matched MLP, and the paired tests show that these gaps are not marginal. This supports the broader argument in the recent dendritic-computing literature that local branch processing can provide useful computational structure rather than merely biological ornamentation [9–13]. In our benchmark, that value appears as a compact representational gain under a small parameter budget rather than as an absolute accuracy gain over large modern vision models.

The stronger scientific question, however, is whether the gain comes from dendritic computation itself or from branching alone. Here the answer is more nuanced. DANN-LRF beats Naive-Branch on FashionMNIST full and CIFAR-10 full, but the margins are much smaller than those against MLP-Param and they disappear on KMNIST and the most reduced Fashion-

MNIST condition. This suggests that structured branching already explains part of the benefit, while dendrite-level nonlinearity contributes an additional but condition-sensitive increment. That distinction matters because adjacent efficiency work has also shown benefits from sparse routing, dendritic normalization, or structured connectivity even without our exact DANN formulation [15–17, 21]. The present Naive-Branch control therefore helps separate two claims that are often conflated in the literature: branching can help, and dendritic nonlinearity may add further benefit, but the second claim must be tested against the first one directly.

Dataset dependence is the second major theme. On FashionMNIST, the local receptive-field prior appears well aligned with the underlying spatial regularity, and DANN-LRF is the strongest low-parameter model. On KMNIST, by contrast, DANN-RANDOM is numerically preferred, while DANN-LRF and Naive-Branch are effectively tied. A plausible interpretation is that KMNIST contains more distributed stroke relationships than the local-patch prior captures well. This is an interpretation rather than a proof of mechanism, but it is consistent with the broader lesson from recent dendritic ANN studies: routing strategy, dendritic interaction type, and task structure interact strongly, so no single dendritic design should be assumed to dominate across settings [37–40]. Our results therefore argue for dendritic architecture search under explicit controls, not for a universal local-receptive-field prior.

The CIFAR-10 results are scientifically useful even though the absolute accuracies are far below modern convolutional or transformer-based benchmarks. In a conventional computer-vision framing, these accuracies would be uncompetitive. In the present framing, however, CIFAR-10 acts as a harder stress test for shallow flattened-input models. Under that stress test, DANN-LRF clearly outperforms MLP-Param and improves over Naive-Branch on the full-data condition. The result therefore supports parameter efficiency, not state-of-the-art recognition. This positioning is important because some recent dendritic studies move directly toward deeper forecasting models, lightweight CNN variants, or hardware-aware spiking systems [38, 45–47, 67]. A controlled shallow benchmark remains valuable precisely because it establishes what the architectural prior contributes before additional depth, convolution, recurrence, or neuromorphic machinery is introduced.

The reduced-dataset diagnostics should be interpreted conservatively. They suggest that DANN-LRF can retain some advantage in moderate reduction settings, especially on CIFAR-10 0.2, but they do not justify a

strong low-training-data robustness claim because the historical archived protocol also affected the test loader. The negative FashionMNIST 0.1 result is therefore important and should remain visible in the paper. It shows that DANN-LRF is not universally favored under more severe data reduction. That negative result also prevents an overly optimistic reading of recent biologically inspired efficiency work [13, 15]: dendritic inductive bias can help, but its benefits remain regime-dependent and should be stated with the same caution as its successes.

Another useful implication concerns architecture versus learning rule. A major branch of biologically plausible learning research studies segregated dendrites, predictive coding, feedback control, and local credit assignment as alternatives or complements to standard backpropagation [51, 55, 58–60]. Our study does not address that learning-rule question. Instead, it shows that even under ordinary supervised optimization, the neuronal architecture itself can inject a measurable inductive bias. This suggests that dendritic architecture and biologically plausible learning rules should be treated as complementary research directions rather than as mutually exclusive explanations. A future system could combine the architectural controls used here with compartment-aware credit-assignment schemes to test whether the two sources of inductive bias are additive.

Finally, the timing analysis clarifies the computational trade-off. DANN-LRF is not free. It is clearly slower than MLP-Param in both training and inference. Yet the additional cost beyond Naive-Branch is small, which means that most of the compute cost comes from routed branching itself, not from adding the dendrite-level nonlinearity on top of that structure. This makes DANN-LRF a plausible candidate when the research goal is to trade modest runtime overhead for higher accuracy under a small parameter budget. That conclusion also aligns with the recent framing of dendrites as efficiency-oriented computational resources rather than as gratuitous biological embellishments [11–13].

6. Limitations and Reproducibility Notes

Several limitations should be stated explicitly.

1. All models operate on flattened image inputs. The study does not compare against convolutional backbones, residual networks, or transformers, and it should not be read as a modern vision benchmark.

2. VANN-Same is a high-capacity reference, not a fair parameter-matched competitor. Its higher absolute accuracy does not invalidate the parameter-efficiency question, but it does bound how much can be claimed about absolute performance.
3. The archived reduced-data folders were generated before the train-subset and test-subset logic were cleanly separated in the repository. They are therefore presented as reduced-dataset diagnostics, not as strict fixed-test low-training-data evidence.
4. The historical summary CSV files export best test accuracy across epochs. To avoid test-set selection bias, the manuscript reconstructs the primary metric from per-epoch histories and uses test accuracy at the best validation epoch instead.
5. The archived accuracy runs were completed across mixed hardware contexts. This does not undermine the accuracy comparisons themselves, but it means that wall-clock interpretation must rely on the dedicated CPU timing protocol rather than on the accuracy folders.
6. The archived result package predates the repository-side fix that now sets all seeds before model construction. The historical outputs remain usable, but this detail slightly weakens strict rerun-level determinism for those archived folders.
7. The CIFAR-10 folders in the archived package evaluate DANN-LRF against controls, but they do not include DANN-RANDOM or DANN-GRF. As a result, the dataset-dependence conclusion about routing strategy is strongest for FashionMNIST and KMNIST.

These limitations do not invalidate the main full-data findings. They instead define the scope of the evidence: controlled, shallow, parameter-aware benchmarking of dendritic architectures, with especially strong support for the full-data comparisons and more cautious support for the archived reduced-data diagnostics.

7. Conclusion

This study asked a narrow but important question: when do dendritic artificial neural networks help once parameter count, branching structure, and runtime are controlled? The answer is favorable but not universal. DANN-LRF consistently improves over a parameter-matched MLP on the full-data benchmarks and on archived CIFAR-10 0.2, while its advantage over naive

branching is smaller and condition-dependent. The preferred routing strategy is dataset-dependent, with local receptive fields favored on FashionMNIST and random routing numerically favored on KMNIST. CPU timing shows that the extra cost of dendritic nonlinearity beyond branching is modest, although DANN-LRF remains slower than the plain MLP baseline. Taken together, the results support DANNs as parameter-efficient classifiers whose value lies in controlled architectural trade-offs rather than in universal accuracy maximization.

CRediT authorship contribution statement

Özkan Canay: Conceptualization, methodology, software, validation, formal analysis, investigation, visualization, writing - original draft, writing - review and editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The codebase, archived historical outputs, manuscript-derivation scripts, and generated tables and figures are prepared for public release at `https://github.com/canay/dendritic-ann-benchmarks`. Dataset access notes and source links for FashionMNIST, KMNIST, and CIFAR-10 are documented in the repository so that reviewers can reproduce downloads from the original dataset providers. The manuscript tables were reconstructed from per-epoch history files to report validation-selected test accuracy as the primary metric.

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Table 5: Seed-paired statistical contrasts for DANN-LRF using validation-selected accuracy. Mean differences are reported in percentage points. Holm correction is applied within each condition.

Condition	**Contrast**	**Mean diff**	**95% CI**	**Holm p**
FashionMNIST full	DANN-LRF − MLP-Param	1.67	[1.54, 1.80]	0.0059
	DANN-LRF − Naive-Branch	0.37	[0.26, 0.48]	0.0059
	DANN-LRF − DANN-RANDOM	0.55	[0.41, 0.69]	0.0059
KMNIST full	DANN-LRF − MLP-Param	4.06	[3.73, 4.37]	0.0059
	DANN-LRF − Naive-Branch	−0.02	[−0.35, 0.31]	0.8457
	DANN-LRF − DANN-RANDOM	−0.51	[−0.91, −0.13]	0.0977
CIFAR-10 full	DANN-LRF − MLP-Param	17.04	[16.06, 18.12]	0.0039
	DANN-LRF − Naive-Branch	1.01	[0.68, 1.34]	0.0039
FashionMNIST 0.2	DANN-LRF − MLP-Param	0.44	[−0.04, 0.97]	0.4922
	DANN-LRF − Naive-Branch	0.13	[−0.29, 0.53]	0.4922
FashionMNIST 0.1	DANN-LRF − MLP-Param	0.01	[−0.43, 0.41]	0.8457
	DANN-LRF − Naive-Branch	−0.04	[−0.30, 0.26]	0.8398
CIFAR-10 0.2	DANN-LRF − MLP-Param	13.18	[12.18, 14.23]	0.0039
	DANN-LRF − Naive-Branch	0.26	[−0.54, 1.11]	0.7344

Table 6: CPU-only timing summary from the dedicated timing runs. Accuracy values are included only as contextual outputs from the timing protocol and are not the main manuscript metric.

Dataset	**Model**	**Params**	**Acc. (%)**	**Train s/epoch**	**Infer ms/1000**
FashionMNIST	DANN-LRF	10,634	85.17	5.926	67.388
	MLP-Param	10,527	84.10	4.043	49.814
	Naive-Branch	10,634	84.58	5.756	66.706
	VANN-Same	468,874	88.09	6.548	59.833
CIFAR-10	DANN-LRF	10,634	43.72	5.280	70.215
	MLP-Param	9,271	28.61	3.891	61.864
	Naive-Branch	10,634	42.82	5.095	67.961
	VANN-Same	1,640,330	49.36	8.041	85.130